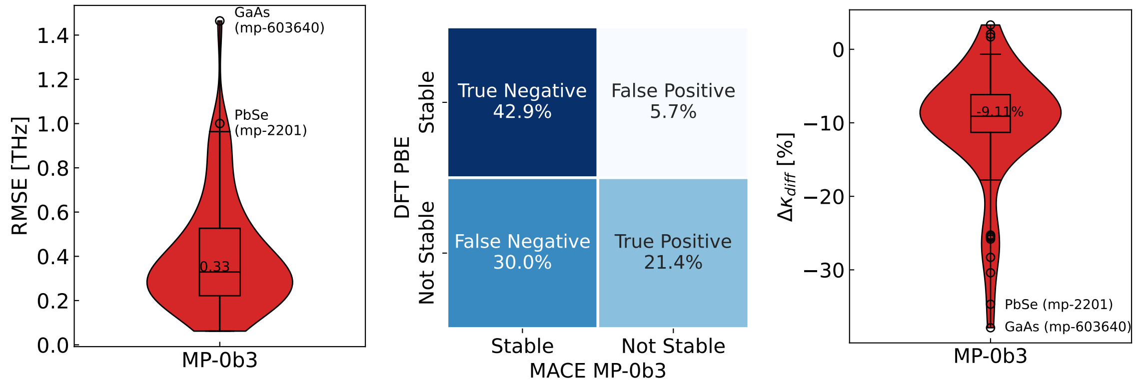

(a) Violin/Box plot of phonon RMSE.

(b) confusion matrix

(c) Relative error of diffusive thermal conductivity

Figure S57: Comparison of DFT PBE and MACE-MP-0b3 phonons. (a) shows the violin and box plot of the RMSE over the whole Brillouin zone. (b) shows the confusion matrix of occurrence of imaginary modes with a threshold of $|\omega_{imag}| < 0.05$ THz. (c) Relative Error of the diffusive thermal conductivity

displacement amplitude to 0.01 Å and setting the minimum edge length to 20 Å, which allows us to perform the calculation with only one k-point. All data and scripts for reproducing the benchmark can be found under: https://doi.org/10.5281/zenodo.15462976

As examples, we show two compounds out of our set where DFT and MACE-MP-0b3 agree well (Figs. S56a and b), and one of the compounds of largest qualitative disagreement (Fig S56c). The noticeable drop in the optical modes around Gamma for PbTe arises from neglecting the non-analytic corrections derived from Born charges, which are unavailable from electronic structure-less ML potentials. The parity plot of MACE-MP-0b3 vs PBE highest-frequency phonon modes in Fig. S56d reveals that MACE-MP-0b3 typically slightly underestimates the highest overall frequency in the phonon band structure (henceforth referred to as bandwidth, which correlates with the stiffness of the material's strongest bond) by a mean absolute error of 0.55 THz in comparison to DFT. MACE-MP-0b3 bandwidths have an excellent predictive power of $R^2 = 0.88$. The least squares regression (red line), with a slope close to one, indicates that the error is largely independent of the actual bandwidth. The largest absolute discrepancy between PBE vs MACE-MP-0b3 maximal phonon frequency is around 2.7 THz (47%) for orthorhombic GaAs (mp-603640). In contrast, 49 out of 70 materials show a maximal phonon frequency error of less than 10%. The violin and box plot of the root mean square errors (RMSEs) in Fig. S57a indicate a median error of 0.33 THz. We calculate the RMSE over the entire Brillouin zone, as the density of states (DOS) is the only invariant quantity and the choice of the k-path is arbitrary. Highly specialised models for one element can achieve much lower errors of 0.1–0.2 THz over the whole band structure (*313*). MACE-MP-0b3 attains this level of accuracy for a few materials but does not achieve it on average. However, the deviations are better compared to results from models that specifically train only on the highest phonon peak on MatBench (*314*). Here, the best models arrive at mean absolute errors of 0.8–1 THz for the last DOS peak of phonons, which is a similar but not identical quantity to the width of the band.

As shown in the confusion matrix inset to Fig. S57, the model also provides a signal on the presence of imaginary modes. The presence of such lattice vibrations indicates that the structure is dynamically unstable, meaning it can transform into a lower-energy structure by displacing atoms along the corresponding vibrational eigenvector. MACE-MP-0b3 achieves 64% accuracy on binary dynamic stability classification with a PBE unstable rate of 51.4% (a nearly balanced data set). Although this outperforms the dummy accuracy of 50%, it also leaves room for significant improvement. A set threshold of 0.05 THz determines how large the absolute value of the imaginary mode must be to count as imaginary. For example, a mode depicted at -0.04 THz in the phonon band structure would not be considered dynamically unstable while -0.06 THz would. The false negative values are strikingly high. We assume there are three reasons for this. Firstly, a caveat to the above analysis is that the PBE unstable labels we use as ground truth may themselves contain false negatives due to potential convergence errors. Secondly, the data set contains mainly thermoelectric materials, which are more prone to the occurrence of imaginary modes. Thirdly, some materials in the dataset have exceptionally long unit cells (20 Å) in one direction. In these nearly amorphous structures in one direction, the energy landscape becomes more complex due to the increased degrees of freedom and leads faster to false positives or false negatives.

As is, MACE-MP-0b3 can already serve as a useful pre-screening filter for dynamically unstable materials, especially given the lower false positive (5.7%) than false negative rate (30.0%). This means MACE-MP-0b3 is less likely to predict materials as unstable that are stable than vice versa and, therefore, is biased toward keeping materials in the candidate pool.

We also emphasize the low computational cost of these predictions. Generating a complete phonon band structure on an Apple M2 Max CPU takes approximately 30 s. Running MACE-MP-0b3 on a supercomputer could easily generate approximate phonon bands for every material in MP or other large databases. However, the accuracy of harmonic phonons of MACE-MP-0b3 is still insufficient to be a standard substitute for DFT.

However, MACE-MP-0b3 is already well suited for estimating diffusive thermal conductivity, which represents a lower limit of thermal conductivity. Following the proposed model of Agne et al. (*315*), the diffusive thermal conductivity in the high-temperature limit only depends on the average phonon frequency. Figure S57 shows the violin and parity plot of the relative diffusive thermal conductivity error of the prediction by MACE-MP-0b3. Since the phonon bandwidth is systematically underestimated (Fig.S56d), a consistent downshift is observed, with a median of -9.11%. The other differences in the phonon band shapes between DFT and MACE-MP-0b3 are effectively averaged out. Imaginary modes were not included in the calculation. The long tail in the violin plot arises from a few cases where the maximum phonon frequencies are significantly underestimated. This occurs due to a softening effect (*265*), which results in an effective underestimation of the PES and, consequently, an underestimation of the forces, leading to a reduced phonon mean frequency.

Similarity statement

The structures whose phonon predictions we analyzed are all part of the training set. However, the supercells, including single-atom perturbations performed by the finite-displacement method are not.

B.2 Bulk and Shear Moduli

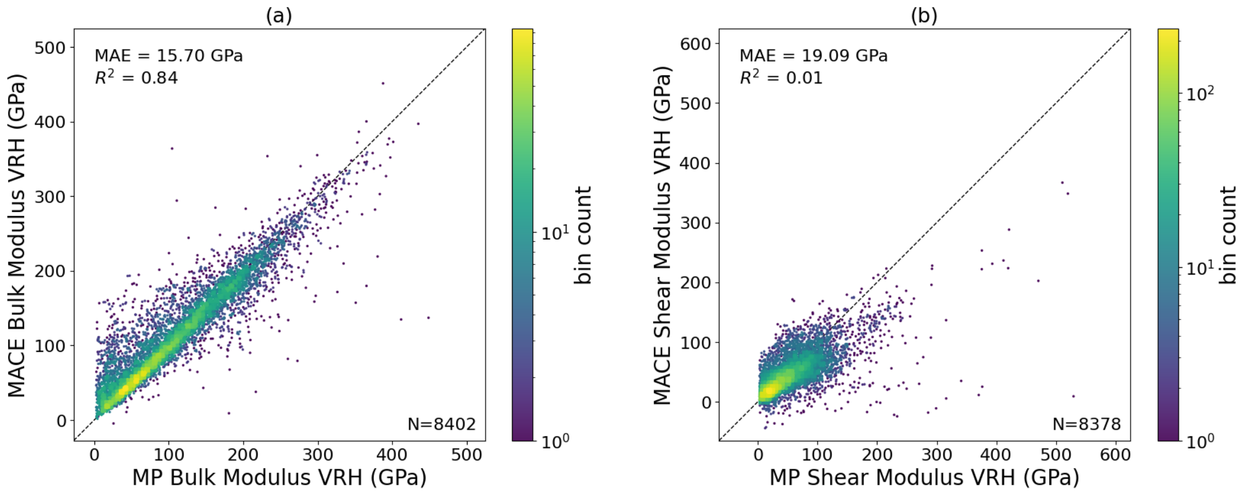

Figure S58: Comparison between MACE-MP-0- and MP-calculated (a) bulk moduli and (b) shear moduli for approximately 8,400 materials stored in the MP Database. The dashed line is a parity line (*i.e.* the target distribution of data). For shear moduli, note that 10 points were excluded from this plot, as MACE-MP-0 predicted an unphysically high ($\geq$600 GPa) or low ($\leq$−50 GPa) shear modulus; this includes mp-{1007819, 1008669, 1009019, 11478, 2458, 27954, 631377, 631633, 721759, 984628}.

MACE-MP-0 was benchmarked against the elastic properties of over 8,000 materials stored in the MP database. Being able to capture elastic properties such as bulk and shear modulus - which depend on the second derivatives of the energy with respect to strain - demonstrates a more precise ability to capture the potential energy surface.

Specifically, MACE-MP-0 was used to calculate the Voigt-Reuss-Hill average (*316–318*) bulk modulus and shear modulus as derived from stress-strain relations. The initial structures used for these calculations were the relaxed PBE (*118*) structures from MP; these structures were then re-relaxed using the MACE-MP-0 model, and then deformed. Specifically, a total of 4 strain magnitudes were used along 6 independent strain modes (in Voigt notation): $\boldsymbol{\epsilon} \in [\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{44}, \epsilon_{55}, \epsilon_{66}]$. For ϵ_{11}, ϵ_{22}, and ϵ_{33}, the strain magnitudes were ± 0.01 and ± 0.005. For ϵ_{44}, ϵ_{55}, and ϵ_{66}, the strain magnitudes were ± 0.06 and ± 0.03. These calculations were performed using the `elasticity` module from the `MatCalc` package (*319*). Hence, all of these predictions are based on equilibrium, bulk crystals alone. Moreover, to filter out likely unphysical DFT predictions, elastic properties from MP were excluded from this analysis if the DFT VRH average bulk or shear modulus are less than −50 GPa or greater than 600 GPa. Note that the data excluded due to unphysical DFT-based properties are distinct from the data not plotted in Fig. S58 due to poor MACE-MP-0 predictions.

Results comparing MP and MACE-MP-0 bulk moduli with MAE of 15.70 GPa and R^2 of 0.84 are shown in Fig. S58(a). This compares favorably to the R^2 value of 0.757 reported for M3GNet (*94*). Similarly, results for shear moduli are shown in Fig. S58(b). MACE-MP-0 struggles to predict shear properties, likely due to a lack of sheared structures in MP.

Similarity statement

None of the DFT deformation calculations contained in MP are present in the MPtrj training set used for MACE-MP-0.

B.3 Cohesive energies

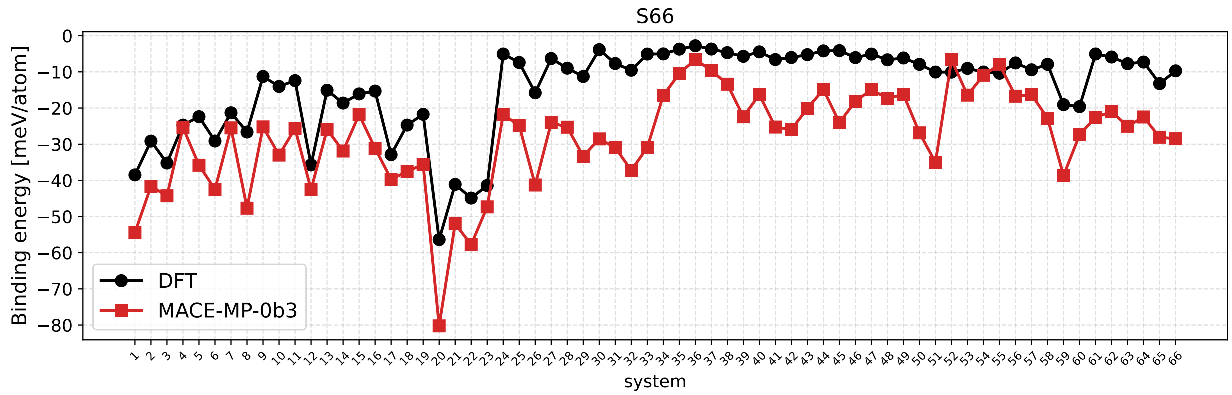

Figure S59: Comparison between DFT (black) and MACE-MP-0 (red) calculated binding energies of the S66 dimers. The binding energies are divided by the number of atoms in each dimer. The lines are guides for the eye.

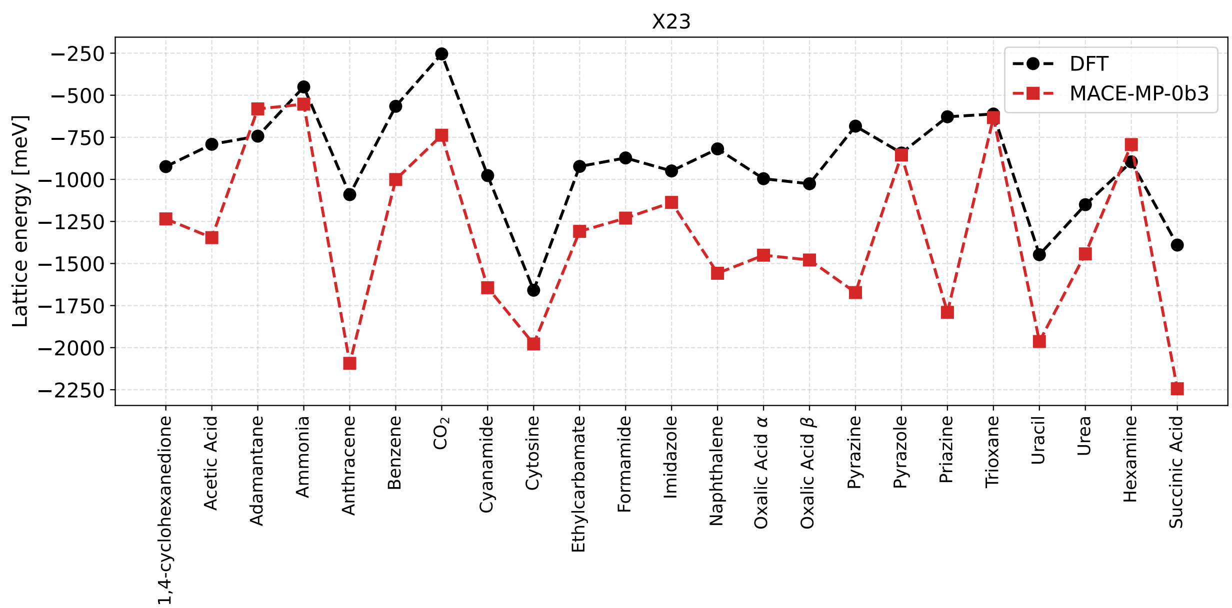

Figure S60: Comparison between DFT (black) and MACE-MP-0-calculated (red) lattice energies of the X23 dataset. The lines are guides for the eye.

In this section, we benchmark MACE-MP-0 against the cohesive energies of widely used data sets of molecules and molecular crystals, S66 (*88*) and X23 (*89*). S66 is a dataset comprising 66 molecular complexes at their reference equilibrium geometries, designed to cover the most common types of noncovalent interactions in biomolecules while keeping a balanced representation of dispersion and electrostatic contributions. X23 is a dataset of 23 organic molecular crystals. Furthermore, we analyze the relative stabilities of the ice polymorphs in DMC-ICE13 (*35*). The DFT calculations were performed using VASP (*4, 295, 296*) with the PBE functional and D3 dispersion correction with Becke-Johnson damping. The energy cutoff is 520 eV. Gas phase calculations were performed at the Γ point in a 25 Å cubic box. Solid phase calculations were performed with a $4 \times 4 \times 4$ k-point grid. The results comparing MACE-MP-0 and DFT binding energies of the S66 dimers, and the lattice energies of X23 and DMC-ICE13 are shown in Fig. S59, Fig. S60, and

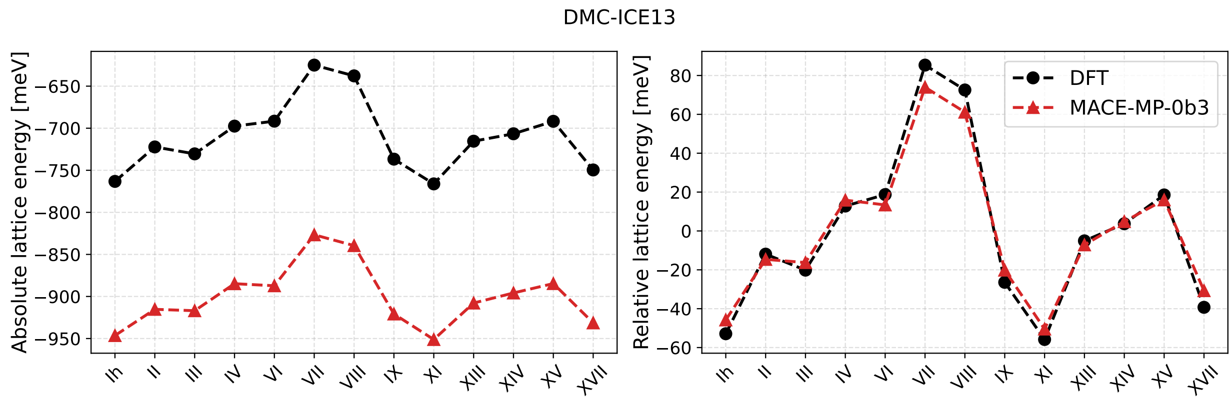

Figure S61: Comparison between DFT (black) and MACE-MP-0b3 (red) lattice energies of the DMC-ICE13 dataset. We report both the absolute lattice energies (left), *i.e.* the energy per molecule of each crystalline phase with respect to the gas phase, and the relative lattice energies (right), *i.e.* the lattice energy relative to mean value among the 13 polymorphs.

Fig. S61. The MAE is approximately 4 meV/atom for S66, 459 meV for X23, and 190 meV for the DMC-ICE13 absolute lattice energies. The relative stabilities of the ice polymorphs are correctly captured with an MAE of 5 meV on the relative lattice energies.

Similarity statement

S66 and X23 comprise dimers and molecular crystals containing C, H, N, or O atoms. Ice polymorphs contain H and O atoms. The MP database contains 73799 structures with O atoms, 10312 structures with H atoms, 11356 structures with N atoms, and 9043 structures with C atoms. The database contains 6 structures matching an exact chemical formula in S66; these are H_4O_2, $C_4HO_8O_4$, C_8H_8 and C_4H_4. 16 structures match an exact chemical formula in X23; these are $C_8H_{16}O_8$, $C_{20}H_{32}$, $H_{12}N_4$, C_4O_8, $C_8H_{16}N_{16}$ and $C_2H_8N_4O_2$. 9 structures match an exact chemical formula in DMC-ICE13; these are $H_{24}O_{12}$ and $H_{16}O_8$. Overall, the database contains 630 structures with organic molecules and 1342 structures with water molecules. We provide `s66_chemiscope_input.json`, `x23_chemiscope_input.json`, and `dmcice13_chemiscope_input.json` to help visualize the interactive UMAP on `chemiscope.org`.

B.4 Atomization energies and lattice constants of solids

In the following section, we benchmark MACE-MP-0 against the atomization energies and lattice constants of a set of solids. The details of the DFT calculations are the same as in appendix B.3. Solid phase DFT total energies were computed with a $16 \times 16 \times 16$ k-point grid. MACE-MP-0 and DFT atomization energies (in eV/atom) are reported in table S5. The MAE is 0.03 eV/atom. Lattice constants are computed on equilibrium structures obtained by geometry relaxation with a force convergence threshold of 0.03 eV/Å. MACE-MP-0 and DFT lattice constants (in Å) are reported in table S6. The MAE is 0.03 Å.

Similarity statement

All the tested solids are contained in the database, with an exact matching chemical formula for 71 structures. We provide `solids_chemiscope_input.json` to help visualize the interactive UMAP on `chemiscope.org`.